Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY and KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID

Rebuttal Testimony

of

Paul M. Normand

Case 19-G-0309
Case 19-G-0310

September 18, 2019

Rebuttal Testimony of Paul M. Normand

TABLE OF CONTENTS

I. Introduction....1

II. Purpose of Testimony....1

III. Response to Staff....2

IV. Response to NYC....7

V. Conclusion....8

I. Introduction

Q. Please state your name, address and business affiliation.

A. My name is Paul M. Normand. I am a Principal with the firm of Management Applications Consulting, Inc. ("MAC"), 1103 Rocky Drive, Suite 201, Reading, Pennsylvania 19609.

Q. Are you the same Paul M. Normand who previously submitted testimony in these proceedings?

A. Yes. The defined terms in my direct testimony have the same definitions here unless otherwise indicated.

II. Purpose of Testimony

Q. What is the purpose of your rebuttal testimony?

A. The purpose of my rebuttal testimony is to address certain recommendations of the Department of Public Service Staff Rates Panel ("Staff") and Lane Kollen on behalf of the City of New York ("NYC") related to the Companies' proposals to modify their leak prone pipe ("LPP") depreciation accrual rates and net salvage factors to provide for the recovery of the costs of the LPP assets and associated net salvage over a 20-year period.

Q. Do you sponsor any exhibits as part of your rebuttal testimony?

A. Yes. I sponsor the following exhibits that were prepared under my direction and supervision:

(1) Exhibit __ (PMN-1R), Staff's response to IR-NG-14; and

(2) Exhibit __ (PMN-2R), which updates Exhibit __ (PMN-4) to include annualized 2019 data for the Companies.

III. Response to Staff

Q. Please summarize Staff's recommendations and their impacts on your depreciation recommendations concerning the Companies' LPP assets.

A. Staff generally agrees that the recovery of costs associated with LPP should be accelerated but recommends the use of a 30-year amortization period instead of the 20-year period proposed by the Companies. Staff further recommends that the net salvage costs reflected in the proposed depreciation rates should be determined based upon a four-year band of net salvage costs incurred by the Companies rather than the three-year band proposed by the Companies. Before the correction of an error that I discuss more fully below, Staff's recommendations would reduce KEDNY and KEDLI's proposed annual recovery of LPP costs by $16.8 million (incorrectly stated as $14.6 million in Staff's testimony) and $10.6 million, respectively, as compared to the Companies' proposals.

Q. Did the Companies identify any errors in Staff's depreciation adjustment for KEDNY?

A. Yes, Staff miscalculated the annual depreciation accrual for KEDNY plant account 380.05 Gas Services – Copper as presented in Exhibit __ (SRP-4) based on Staff's recommendation of a 30-year average service life and a four-year band of net salvage costs. Additionally, Staff included the incorrect depreciation rate for KEDNY's 380.05

plant account in Staff's plant forecast model that ultimately was included in Staff's revenue requirement presented in Exhibit __ (SRRP-2). The specific adjustments in Exhibit __ (SRRP-2), Schedule 7C for Staff's depreciation recommendation that are incorrect are Utility Plant Adjustment 23(b)(2) and Depreciation Adjustment 19(2).

Q. What is the impact of correcting these errors on the revenue requirement?

A. Staff overstated its depreciation reduction adjustment by approximately $3 million, which led to a corresponding overstatement to Utility Plant.

Q. Did Staff agree with the correction noted by the Company?

A. Yes, in Staff's response to IR NG-14 included as Exhibit __ (PMN-1R), Staff acknowledged the error and provided an updated plant forecast model that reflected the correction.

Q. Do you agree with Staff's proposed changes to the Companies' depreciation parameters and rates?

A. No, I do not.

Q. Please discuss the bases of your disagreements with Staff's depreciation recommendations.

A. Although Staff asserts (at 23) that it is "ultimately" concerned with the bill impacts of the Companies' proposals, Staff offers no explanation as to why the impacts of those proposals are unreasonable in these cases. It is my understanding that Staff is

recommending a decrease in base rates for KEDLI of $51.852 million annually and an increase of only $2.205 million annually for KEDNY. Given the size of the proposed base rate increases in this case, even as proposed by KEDNY and KEDLI, it is not clear why Staff believes that the impact of the Companies' LPP depreciation proposals is unreasonable or excessive. To the extent that Staff is concerned about the impact of the Companies' LPP proposals in future rate cases, the Commission can always revisit the Companies' LPP depreciation rates in future cases if the Companies' seek rate increases that have bill impacts that the Commission determines to be unreasonable.

Q. Staff asserts (at 23-24) that it is making the same proposals for KEDNY and KEDLI for consistency. Do you agree that there is need for consistent LPP depreciation parameters?

A. No, I do not. Depreciation rates are typically set on the basis of individual utilities' facts and circumstances. There is no reason why KEDNY and KEDLI should be required to amortize LPP costs in consistent fashion. In this instance, geography alone dictates that a difference can be expected because of the urban nature of KEDNY's service territory and the more suburban/rural nature of KEDLI's territory.

Q. Staff claims (at 24) that it is equitable for current and future customers to pay the costs of the replacement of LPP because it is being done for safety and reliability reasons. Do you agree?

A. No. A significant amount of the facilities that a gas distribution system replaces is replaced due to safety and reliability concerns. The existence of such concerns does not

make it more equitable for the Commission to require future customers to pay for both facilities that are replaced as well as the new facilities that replace them. The Companies and Staff agree that the Companies' aging LPP infrastructure should be replaced over 20 years. The recovery of the costs of the replaced facilities should be in sync with the same period of time. This will result in an equitable cost recovery from all customers with minimal cost shifting from one population of customers to another (future) population. This is particularly important because the cost of removing LPP facilities, which is primarily labor expense, will rise over the period in which the LPP is replaced and the Companies' are using current average historical costs to estimate the cost removal in future years. For this reason alone, using the current proposed cost of removal is understated and future customers will likely be required to pay more. In addition, it is more important than ever for the Companies to secure an equitable level of LPP cost recovery from current customers because there is a likelihood that the Companies' customer counts may drop in the future as a result of political decisions to reduce the use of fossil fuels. Passing cost responsibility for the cost of removing LPP assets that are being replaced today onto a diminishing number of future customers would be particularly unfair. Staff's recommendation to extend responsibility for the cost of LPP assets for a period of 10 years beyond the time in which those assets will be replaced does not result in a fair and equitable assignment of costs.

Q. Staff asserts (at 24-25) that the use of a four-year historical analysis of net salvage is more reasonable than the Companies' proposed three-year analysis. Do you agree?

A. No. Staff's proposal looks further back in time to reflect a lower cost of removal. However, there is no reason to believe that the cost of removal is likely to decrease in the future. To the contrary, this cost is more likely to increase because the cost is driven primarily by labor costs that will continue to increase over time. This is driven by both the impact of general inflation and what is likely to be a tightened market for the skilled labor needed to replace utility facilities. Staff's use of data from 2015 to reduce net salvage costs makes it more likely that the Company will underrecover its net salvage costs.

The analysis presented in my direct testimony focused on looking at a span of years that would capture annual fluctuations in reported cost of removal data as well as the most recent data to capture as fully as possible the costs of the ongoing replacement programs. The 2015 cost of removal data that Staff seeks to include was omitted because costs in the first year of a 20-year replacement program are less likely to be representative of the ongoing costs of a mature program.

As with all these types of projects, time is required to ramp up the annual activity of a 20-year program. My review started with the KEDNY data as shown on Exhibit __ (PMN-4), pages 3 and 4 that presents annual costs relating to retirements and cost to retire. In reviewing this data, I observed that the annual fluctuations were considerable, which is not unusual because, in many instances, there is a lag in the reporting of data for different years. As a result of this data lag, I suggested that a span of several years would be appropriate. I chose three years as the most representative over four years as

recommended by Staff because adding the fourth year showed some high potential dollar lags that materially affected the results. Given the fact that the cost of replacing LPP is likely to rise over time, I determined that 2015 reported data should not be included in the analysis because it would reduce the cost based on a set of start-up circumstances that were unlikely to recur. Adding 2015 data to the analysis simply dilutes the cost estimates.

To validate the reasonableness of my approach, I prepared an update to the analysis that I originally included as Exhibit __ (PMN-4). That update is included as Exhibit __ (PMN-2R) and reflects the addition of actual retirement data for the first eight months (January – August) of calendar year 2019 and annualizes that data for the full 12 months. Data is provided for KEDNY on page 1 and KEDLI on page 2 and the annualization of the data is set forth on page 3. A comparison of the 2019 data to the earlier years clearly shows that the proposed three-year average is the most appropriate data to use to estimate the cost of removal going forward and that the 2015 data is an outlier.

IV. Response to NYC

Q. What is your understanding of Mr. Kollen's recommendation concerning the Companies' LPP depreciation proposals?

A. Mr. Kollen asserts that the Companies' proposals to accelerate the recovery of LPP assets should be rejected. He further asserts that the Companies should be required to defer their actual costs of removal for future recovery and then recover the actual costs through existing depreciation rates.

Q. Do you agree with Mr. Kollen's recommendations?

A. No. Mr. Kollen's recommendations are nothing more than an exercise in kicking the can down the road so that future customers will be required to bear significant portions of the costs of LPP assets that are being retired and the costs of removing those facilities as well as the costs of the replacement facilities. His proposal would negatively affect the Companies' cash flow and is completely unfair to future customers.

Q. Do you agree with Mr. Kollen's concerns about the reliability of your estimates of cost of removal?

A. No, I do not. Using a multi-year average of costs of removal mitigates concerns about the reliability of the estimates. Moreover, the increased cost of labor that is likely to be incurred in the future, which is the bulk of cost to be recovered, is not captured by the analysis. Thus, there is little risk that the cost of removal that the rates will recover is likely to be less than the actual cost.

V. Conclusion

Q. Does this conclude your rebuttal testimony?

A. Yes.

INDEX OF EXHIBITS

Exhibit____(PMN-1R): Staff's response to IR-NG-14

Exhibit____(PMN-2R): Updated Exhibit __ (PMN-4) to include annualized 2019 data for the Companies

Exhibit ___ (PMN-1R)

Staff's response to IR-NG-14

STAFF OF THE DEPARTMENT OF PUBLIC SERVICE
RESPONSE TO INTERROGATORY/DOCUMENT REQUEST

The Brooklyn Union Gas Company d/b/a National Grid NY
KeySpan Gas East Corporation d/b/a National Grid
Cases 19-G-0309 & 19-G-0310 – Gas Rates

Request No.:	NG-14
Requested By:	Paul Normand and Revenue Requirement Panel
Information Requested of:	Staff Rates Panel
Date of Request:	September 3, 2019
Response Due Date:	September 13, 2019
Date of Response:	September 13, 2019
Name & Position of Respondent:	Mingdi Huang, Utility Engineering Specialist 1
Subject:	Exhibit___(SRP-4) LPP Depreciation

Question:

1. Please provide Exhibit SRP-4 and all supporting calculations in original electronic excel format with all formulae intact and unlocked.

Response:

1. See Response to IR NG-14 (Attachment). This is an updated version of Exhibit___(SRP-4), which includes the corrections discussed in the questions and responses below.

Question:

2. Please explain why Staff Recommendation for KEDNY Account 380.05 of an annual accrual of $2,686,774 included in Exhibit SRP-4 does not agree with the total $533,041 ($5,441 + $527,600) annual accrual reflected in the "Deprec Rates" tab of the Plant Forecast Model supporting Exhibit SGIOP-5 and Exhibit SGIOP-6.

Response:

2. The Staff Recommendation for the annual accrual of KEDNY Account 380.05 of $2,686,774 does not agree with the total annual accrual of $533,041 reflected in the Plant Forecast Model because the Plant Forecast Model utilized an incorrect depreciation rate of 1.73%. The correct depreciation rate is 11.62%.

STAFF OF THE DEPARTMENT OF PUBLIC SERVICE
RESPONSE TO INTERROGATORY/DOCUMENT REQUEST

The Brooklyn Union Gas Company d/b/a National Grid NY
KeySpan Gas East Corporation d/b/a National Grid
Cases 19-G-0309 & 19-G-0310 – Gas Rates

Response to IR NG-14 continued.

Question:

3. Please confirm that Staff's KEDNY Depreciation Adjustment #19(2) in SRRP-2 for a reduction of $16.784 million was calculated by comparing the Company's LPP depreciation proposal of $29.258 million to the $12.474 million included in Staff's Plant Forecast Model that included only $533,041 of depreciation for Account 380.05.

Response:

3. Yes, the reduction of $16.784 million was calculated by comparing the Company's LPP depreciation proposal of $29.258 million to the $12.474 million included in Staff's Plant Forecast Model.

Question:

4. Please confirm and explain KEDNY's Exhibit SRP-4 for Account 380.05 presents $2,686,744 annual depreciation accrual calculated using a 40-year average service life versus the 30-year average service life used for all other accounts with leak prone pipe assets.

Response:

4. Yes, the annual depreciation accrual of $2,686,744 was incorrectly calculated using a 40-year average service life. The Response to IR NG-14 (Attachment) reflects a 30-year average life.

Question:

5. Please explain why Staff did not consistently use a 30-year average service life for all accounts as described in the Staff Rates Panel testimony.

Response:

5. The use of a 40-year average service life was incorrect. Staff's recommendation is to use a 30-year average service life for all accounts, as described in the Staff Rates Panel testimony.

STAFF OF THE DEPARTMENT OF PUBLIC SERVICE
RESPONSE TO INTERROGATORY/DOCUMENT REQUEST

The Brooklyn Union Gas Company d/b/a National Grid NY
KeySpan Gas East Corporation d/b/a National Grid
Cases 19-G-0309 & 19-G-0310 – Gas Rates

Response to IR NG-14 continued.

Question:

6. Please confirm that if Staff used a 30-year average service life for KEDNY Account 380.05 the annual accrual would be $3,580,311 vs. the $2,686,744 presented in SRP-4 and the $533,041 included in Staff's Plant Forecast Model.

Response:

6. Yes, if a 30-year average service life was used in the calculation, the annual accrual would equal $3,580,311 for KEDNY Account 380.05. The Response to IR NG-14 (Attachment) reflects a 30-year average life.

Question:

7. If the $2,686,744 was in error and Staff agrees with the $3,580,311 calculation, please confirm that Staff's annual accrual for KEDNY is understated in the Plant Forecast Model by $3,047,270 (i.e. $3,580,311 - $533,041). If not, explain why.

Response:

7. Yes, the annual accrual for KEDNY is understated by $3,047,270 in the Plant Forecast Model.

		Current							Company Proposed								Staff Proposed							
Account #	Description	Plant Balance	Average Service Life	Average Remaining Life	Average Consumed Life	Net Salvage	Annual Depreciation Accrual Rate	Annual Depreciation Accrual	Book Reserve	Average Service Life	Net Salvage	Net Salvage Factor	Annual Depreciation Accrual Rate	Annual Depreciation Accrual	Theoretical Reserve without Net Salvage	Theoretical Reserve With Net Salvage	Difference Between Proposed and Current Annual Accrual	Reserve Variance	Average Service Life	Net Salvage	Net Salvage Factor	Annual Depreciation Accrual Rate	Annual Depreciation Accrual	Theoretical Reserve without Net Salvage	Theoretical Reserve With Net Salvage	Difference Between Proposed and Current Annual Accrual	Reserve Variance
367.08	MAINS-STEEL	731,258	80	28.01	51.99	-60	1.25	9,141	591,013	20	-357.3	4.573	22.87	167,239	475,191	2,173,048	158,098	1,582,035	30	-274.04	3.7404	12.47	91,188	475,191	1,777,404.42	82,047	1,186,391
376.05	MAINS-CAST IRON	15,314,354	50	9.51	10.49	-18	2.76	422,676	(47,915,755)	20	-1968.1	20.681	103.41	15,836,573	12,401,719	254,479,951	15,413,897	304,195,706	30	-1357.58	14.5758	48.59	7,441,245	12,401,719	180,764,976	7,018,569	228,680,731
376.09	MAINS-STEEL	30,434,655	75	29.98	45.02	-65	2.2	669,562	16,706,814	20	-357.3	4.573	22.87	6,960,406	18,269,976	83,548,630	6,290,844	66,841,766	30	-274.04	3.7404	12.47	3,795,201	18,269,976	68,337,018	3,125,639	51,610,184
380.02	SERVICES-STEEL	8,682,297	55	25.18	29.82	-5	1.91	165,832	8,301,302	20	-111.9	2.119	10.6	920,323	4,706,707	9,973,512	754,491	1,672,210	30	-111.88	2.1188	7.06	612,970	4,706,707	9,972,571	447,138	1,671,269
380.05	SERVICES-COPPER	30,811,627	60	29.84	30.16	-4	1.73	532,852	15,948,718	20	-248.7	3.487	17.44	5,373,548	15,484,930	54,009,899	4,840,696	18,061,161	30	-248.72	3.4872	11.62	3,580,311	15,484,930	54,012,997	3,047,459	18,064,259
				24.504	33.496			1,800,063	13,632,132					29,258,089		406,185,010	27,458,026	392,552,878					15,520,915		314,864,966	13,720,852	301,232,834
														13,737,174													(91,320,044)
380.03	SERVICES-COPPER POST '91	314,516				0	1.67	5,252		20			17.44	54,852			49,600										
380.05	SERVICES-COPPER PRE '92	30,497,111				-4	1.73	527,600		20			17.44	5,318,696			4,791,096										
		314,516								30			11.62	36,547			36,547										
		30,497,111								30			11.62	3,543,764			3,543,764										

Current

Account #	Description	Plant Balance	Average Service Life	Average Remaining Life	Average Current Life	Net Salvage	Annual Depreciation Accrual Rate	Annual Depreciation Accrual	Book Reserve
367.04	MAINS-CAST IRON	253,190	50	6.44	41.56	0	2	5,034	(41,921)
367.09	MAINS-STEEL	4,862,146	80	31.87	48.13	0	1.25	60,777	4,123,009
370.00	MAINS-STEEL	47,714,469	80	33.97	46.03	-3	1.29	635,241	33,911,614
376.10	MAINS-CAST IRON	4,038,864	50	20.95	29.05	-14	2.28	92,086	(2,272,645)
376.18	MAINS-PLASTIC	34,841,811	75	38.21	36.79	0	1.33	463,396	19,961,165
378.00	MAINS-STEEL	2,700,183	60	31.41	28.59	0	1.67	301,144	9,139,617
380.06	SERVICES-PLASTIC	31,418,577	60	31.59	28.41	0	1.67	524,690	11,134,217
380.03	SERVICES-COPPER	1,542,826	60	26.4	33.6	0	1.67	25,765	675,897
				25.605	37.10875			2,078,323	

Company Proposed

Average Service Life	Net Salvage	Net Salvage Factor	Annual Depreciation Accrual Rate	Annual Depreciation Accrual	Theoretical Reserve without Net Salvage	Theoretical Reserve With Net Salvage	Difference Between Proposed and Current Accruals	Reserve Variance
20	-2608	3.008	18.04	45,335	238,847	789,600	40,291	831,522
20	-68	7.01	35.05	1,704,182	2,295,590	20,506,984	1,644,405	16,383,975
20	-68	7.01	35.05	16,650,450	18,945,041	214,116,811	16,013,209	197,478,527
20	-2608	3.008	18.04	728,611	2,346,788	8,467,211	636,525	10,739,856
20	-38	1.38	6.9	2,404,085	17,091,305	21,986,001	1,940,689	3,634,836
20	-1412	1.412	17.06	1,340,663	13,143,969	14,484,632	1,340,689	31,410,960
20	-38	1.38	6.9	2,367,882	14,875,159	20,527,719	1,643,192	9,193,502
20	-307	1.307	6.54	300,901	856,078	1,329,350	453,453	
				26,555,139			24,479,866	
				10,623,832				

Staff Proposed

Average Service Life	Net Salvage	Net Salvage Factor	Annual Depreciation Accrual Rate	Annual Depreciation Accrual	Theoretical Reserve without Net Salvage	Theoretical Reserve With Net Salvage	Difference Between Proposed and Current Accruals	Reserve Variance
30	-298.4	3.984	13.18	33,107	218,847	865,321		26,083
30	-519.53	6.8953	20.65	1,004,033	1,583,390	19,123,669		941,256
30	-519.53	6.8953	20.65	13,966,010	16,440,454	198,514,958		13,998,600
30	-298.4	3.984	13.18	532,232	2,346,788	9,279,300		440,226
30	-36.52	1.3652	4.55	1,585,302	17,091,305	23,333,050		1,121,906
30	-36.52	1.3652	4.55	1,143,456	13,143,969	17,941,998		1,557,366
30	-36.52	1.3652	4.55	1,429,545	14,875,159	20,307,767		904,855
30	-27.21	1.2721	4.24	65,416	856,078	1,099,394		39,651
				35,031,139				19,853,014
				13,853,014				

Account #	Description	Difference Between Proposed and Current Accruals	Reserve Variance
367.04	MAINS-CAST IRON	26,083	907,243
367.09	MAINS-STEEL	941,256	13,998,600
370.00	MAINS-STEEL	8,862,967	172,611,418
376.10	MAINS-CAST IRON	440,226	11,551,845
376.18	MAINS-PLASTIC	1,121,906	3,171,885
378.00	MAINS-STEEL	1,557,366	31,966,726
380.06	SERVICES-PLASTIC	904,855	8,973,360
380.03	SERVICES-COPPER	39,651	423,297
		13,853,014	

KEDNY

ACCOUNT 376.05 Mains - Cast Iron	2018	2017	2016	2015	2014	Three-Year Band	COR$/Mile	Four-Year Band	COR$/Mile	Five-Year Band	COR$/Mile
Cost to Retire	$ 22,887,532.95	$ 4,462,616.35	$ 6,990,138.41	$ 1,239,814.41	$ 1,794,411.95	$ 34,340,287.71		$ 35,580,102.12		$ 37,374,514.07	
LPP Miles	78.04	59.19	4.05	84.62	0.82	$ 141.28		$ 225.90		226.72	
							$ 243,065.46		$ 157,503.77		$ 164,848.77
ACCOUNT 376.09 Mains - Steel											
	2018	2017	2016	2015	2014						
Cost to Retire	$ 9,497,923.07	$ 563,284.83	$ 2,665,628.81	$ 1,812,939.99	$ -	$ 12,726,836.71		$ 14,539,776.70		$ 14,539,776.70	
LPP Miles	23.38	6.64	0.41	0.95	-	$ 30.43		$ 31.38		31.38	
							$ 418,233.21		$ 463,345.34		$ 463,345.34
ACCOUNT 380.02 Services - Steel											
	2018	2017	2016	2015	2014						
Cost to Retire	$ 6,804,809.69	$ 182,804.05	$ (48,949.89)	$ -	$ 315,065.63	$ 6,938,663.85		$ 6,938,663.85		$ 7,253,729.48	
LPP # of Services	7,391.00	2,923.00	7,384.00	2.00	54.00	$ 17,698.00		$ 17,700.00		17,754.00	
							$ 392.06		$ 392.01		$ 408.57
ACCOUNT 380.05 Services - Copper											
	2018	2017	2016	2015	2014						
Cost to Retire	$ 267,571.22	$ 188,218.91	$ -	$ -	$ -	$ 455,790.13		$ 455,790.13		$ 455,790.13	
LPP # of Services	255.00	369.00	-	-	-	$ 624.00		$ 624.00		624.00	
							$ 730.43		$ 730.43		$ 730.43

KEDLI

ACCOUNT 367.09 Mains - Steel	2018	2017	2016	2015	2014	Three-Year Band	COR$/Mile	Four-Year Band	COR$/Mile	Five-Year Band	COR$/Mile
Cost to Retire	$ -	$ -	$ -	$ -	$ 883.00	$ -		$ -		$ 883.00	
LPP Miles	-	-	-	-	-	$ -		$ -		-	
							#DIV/0!		#DIV/0!		#DIV/0!
ACCOUNT 376.08 Mains - Steel											
	2018	2017	2016	2015	2014						
Cost to Retire	$ 18,727,922.32	$ 3,437,622.06	$ 3,303,664.73	$ 1,914,246.79	$ 462,318.51	$ 25,469,209.12		$ 27,383,455.90		$ 27,845,774.41	
LPP Miles	106.81	102.73	68.43	63.17	40.79	$ 277.97		$ 341.14		381.93	
							$ 91,625.75		$ 80,270.43		$ 72,908.06
ACCOUNT 376.10 Mains - Cast Iron											
	2018	2017	2016	2015	2014						
Cost to Retire	$ 336,650.07	$ 7,681.94	$ 707,924.47	$ 295,914.17	$ -	$ 1,052,256.48		$ 1,348,170.65		$ 1,348,170.65	
LPP Miles	8.31	-	1.68	5.83	1.16	$ 9.99		$ 15.82		16.98	
							$ 105,330.98		$ 85,219.38		$ 79,397.56
ACCOUNT 376.18 Mains - Plastic											
	2018	2017	2016	2015	2014						
Cost to Retire	$ 64,227.78	$ 178,075.49	$ 68,863.59	$ 227,152.49	$ 178,336.27	$ 311,166.87		$ 538,319.36		$ 716,655.63	
LPP Miles	0.90	0.01	0.10	0.12	1.15	$ 1.01		$ 1.13		2.28	
							$ 308,086.01		$ 476,388.81		$ 314,322.65
ACCOUNT 380.06 Services - Plastic											
	2018	2017	2016	2015	2014						
Cost to Retire	$ 475,097.61	$ 176,015.81	$ 735,160.83	$ 301,346.57	$ 778,234.10	$ 1,386,274.26		$ 1,687,620.82		$ 2,465,854.92	
LPP # of Services	1,518.00	2,144.00	2,375.00	1,602.00	2,140.00	$ 6,037.00		$ 7,639.00		9,779.00	
							$ 229.63		$ 220.92		$ 252.16
ACCOUNT 380.01 Services - Steel											
	2018	2017	2016	2015	2014						
Cost to Retire	$ 1,300,072.08	$ 1,797,309.46	$ 1,641,592.58	$ 472,373.72	$ 4,604,947.93	$ 4,738,974.12		$ 5,211,347.84		$ 9,816,295.77	
LPP # of Services	1,810.00	3,474.00	2,229.00	3,641.01	5,247.00	$ 7,513.00		$ 11,154.01		16,401.01	
							$ 630.77		$ 467.22		$ 598.52
ACCOUNT 380.03 Services - Copper											
	2018	2017	2016	2015	2014						
Cost to Retire	$ 12,137.97	$ 10,154.43	$ 2,371.86	$ 297.36	$ 14,743.96	$ 24,664.26		$ 24,961.62		$ 39,705.58	
LPP # of Services	64.00	71.00	80.00	30.00	84.00	$ 215.00		$ 245.00		329.00	
							$ 114.72		$ 101.88		$ 120.69

KEDNY

			Gross Plant	5-Year Band COR$/Mile	Average Annual Program Miles	LPP Years	Program Annual COR$	Annual COR %	COR% over Life of Asset
ACCOUNT	367.08	Mains - Steel	$ 731,258	$ 463,345.34	9	20	$ (4,170,108.04)	-570.26%	-11405.30%
ACCOUNT	376.05	Mains - Cast Iron	$ 15,314,354	$ 157,503.77	66	20	$ (10,395,248.96)	-67.88%	-1357.58%
ACCOUNT	376.09	Mains - Steel	$ 30,434,655	$ 463,345.34	9	20	$ (4,170,108.04)	-13.70%	-274.04%
ACCOUNT	380.02	Services Mains - Steel	$ 8,682,297	$ 392.01	1,239	20	$ (485,706.47)	-5.59%	-111.88%
ACCOUNT	380.05	Services - Copper	$ 30,811,626	$ 730.43	5,249	20	$ (3,834,042.30)	-12.44%	-248.87%

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			Gross Plant	5-Year Band COR$/Mile	Average Annual Program Miles	LPP Years	Program Annual COR$	Annual COR %	COR% over Life of Asset
ACCOUNT	367.04	Mains - Cast Iron	$ 251,190.00	$ 85,219.38	7	20	$ (596,535.69)	-237.48%	-4749.68%
ACCOUNT	367.09	Mains - Steel	$ 4,862,146.00	$ 80,270.43	148	20	$ (11,880,024.25)	-244.34%	-4886.74%
ACCOUNT	376.08	Mains - Steel	$ 45,733,409.00	$ 80,270.43	148	20	$ (11,880,024.25)	-25.98%	-519.53%
ACCOUNT	376.10	Mains - Cast Iron	$ 4,038,864.00	$ 85,219.38	7	20	$ (596,535.69)	-14.77%	-295.40%
ACCOUNT	376.18	Mains - Plastic	$ 34,841,811.00	$ 101.88	2,597	20	$ (264,593.17)	-0.76%	-15.19%
ACCOUNT	380.06	Services Mains - Plastic	$ 31,418,577.00	$ 220.92	2,597	20	$ (573,733.64)	-1.83%	-36.52%
ACCOUNT	380.01	Services - Steel	$ 19,781,083.00	$ 467.22	3,780	20	$ (1,766,081.87)	-8.93%	-178.56%
ACCOUNT	380.03	Services - Copper	$ 1,542,826.00	$ 101.88	206	20	$ (20,988.14)	-1.36%	-27.21%

The Brooklyn Union Gas Company d/b/a National Grid NY

LPP Depreciation Adjustments

Account #	Description	Plant Balance	Book Reserve	Average Service Life (Years)			Net Salvage			Annual Depreciation Accrual			Theoretical Reserve With Net Salvage	
				Current	Company Proposed	Staff Recommended	Current	Company Proposed	Staff Recommended	Current	Company Proposed	Staff Recommended	Company Proposed	Staff Recommended
367.08	MAINS-STEEL	$ 731,258	$ 591,013	80	20	30	-60%	-357%	-274%	$ 9,141	$ 167,239	$ 91,188	$ 2,173,048	$ 1,777,404
376.05	MAINS-CAST IRON	$ 15,314,354	$ (47,915,755)	50	20	30	-38%	-1968%	-1358%	$ 422,676	$ 15,836,573	$ 7,441,245	$ 256,479,951	$ 180,764,976
376.09	MAINS-STEEL	$ 30,434,655	$ 16,706,834	75	20	30	-65%	-357%	-274%	$ 669,562	$ 6,960,406	$ 3,795,201	$ 83,548,600	$ 68,337,018
380.02	SERVICES-STEEL	$ 8,682,297	$ 8,301,302	55	20	30	-5%	-112%	-112%	$ 165,832	$ 920,323	$ 612,970	$ 9,973,512	$ 9,972,571
380.05	SERVICES-COPPER	$ 30,811,627	$ 35,948,738	60	20	30	-4%	-249%	-249%	$ 532,852	$ 5,373,548	$ 3,580,311	$ 54,009,899	$ 54,012,997
	Total	$ 85,974,191	$ 13,632,132								$ 29,258,089	$ 15,520,915		

KeySpan Gas East Corporation d/b/a National Grid
LPP Depreciation Adjustments

Account #	Description	Plant Balance	Book Reserve	Average Service Life (Years)			Net Salvage			Annual Depreciation Accrual			Theoretical Reserve With Net Salvage	
				Current	Company Proposed	Staff Recommended	Current	Company Proposed	Staff Recommended	Current	Company Proposed	Staff Recommended	Company Proposed	Staff Recommended
367.04	MAINS CAST IRON	$ 251,190	$ (41,922)	50	20	30	0%	-261%	-295%	$ 5,024	$ 45,315	$ 33,107	$ 789,600	$ 865,321
367.09	MAINS-STEEL	$ 4,862,146	$ 4,125,009	80	20	30	0%	-601%	-520%	$ 60,777	$ 1,704,182	$ 1,004,033	$ 20,506,984	$ 18,123,669
376.08	MAINS-STEEL	$ 45,733,409	$ 18,932,624	80	20	30	-1%	-601%	-520%	$ 576,241	$ 16,029,560	$ 9,443,949	$ 216,510,881	$ 191,348,054
376.10	MAINS-CAST IRON	$ 4,038,864	$ (2,272,645)	50	20	30	-14%	-261%	-295%	$ 92,086	$ 728,611	$ 532,322	$ 8,467,211	$ 9,279,200
376.18	MAINS-PLASTIC	$ 34,841,811	$ 19,961,165	75	20	30	0%	-38%	-37%	$ 463,396	$ 2,404,085	$ 1,585,302	$ 23,586,001	$ 23,333,050
380.01	SERVICES-STEEL	$ 19,781,083	$ 9,738,622	60	20	30	0%	-241%	-179%	$ 330,344	$ 3,374,653	$ 1,837,663	$ 41,161,682	$ 33,604,918
380.06	SERVICES-PLASTIC	$ 31,418,577	$ 11,334,217	60	20	30	0%	-38%	-37%	$ 524,690	$ 2,167,882	$ 1,429,545	$ 20,527,719	$ 20,307,567
380.03	SERVICES-COPPER	$ 1,542,826	$ 675,897	60	20	30	0%	-31%	-27%	$ 25,765	$ 100,901	$ 65,416	$ 1,129,350	$ 1,099,194
	Total	$ 142,469,906	$ 62,452,967								$ 26,555,189	$ 15,931,337		

Exhibit ___ (PMN-2R)

Updated Exhibit __ (PMN-4)

NATIONAL GRID - KEDNY
LPP PROGRAM
SALVAGE AND COST OF REMOVAL ANALYSIS

ACCOUNT 376.05 **Mains - Cast Iron**

			YEAR							3 yr. band Total 2016-2018	4 yr. band Total 2015-2018	7 yr. band Total 2012-2018
	2019 - Projected Full Yr	2019 - as of Aug 31st	2018	2017	2016	2015	2014	2013	2012			
Retirements	**477,436**	**318,291**	672,585	555,399	20,094	657,905	3,712	14,975	181,745	1,248,078	1,905,983	2,106,416
Gross Salvage	**0**	**0**	0	0	0	0	0	0	0	0	0	0
Cost to Retire	**23,803,733**	**15,869,155**	22,887,533	4,462,616	6,990,138	1,239,814	1,794,412	2,570,252	10,671,774	34,340,288	35,580,102	50,616,541
Net Salvage$	**-23,803,733**	**-15,869,155**	-22,887,533	-4,462,616	-6,990,138	-1,239,814	-1,794,412	-2,570,252	-10,671,774	-34,340,288	-35,580,102	-50,616,541
Net Salvage %	**-4986**	**-4986**	-3403	-803	-34788	-188	-48343	-17163	-5872	-2751	-1867	-2403

ACCOUNT 376.09 **Mains - Steel**

			YEAR							3 yr. band Total 2016-2018	4 yr. band Total 2015-2018	7 yr. band Total 2012-2018
	2019 - Projected Full Yr	2019 - as of Aug 31st	2018	2017	2016	2015	2014	2013	2012			
Retirements	**462,708**	**308,472**	909,583	205,450	3,728	62,458	0	175,458	416,596	1,118,761	1,181,220	1,773,273
Gross Salvage	**0**	**0**	0	0	0	0	0	0	0	0	0	0
Cost to Retire	**8,932,227**	**5,954,818**	9,497,923	563,285	2,665,629	1,812,940	0	1,606,944	6,757,999	12,726,837	14,539,777	22,904,720
Net Salvage$	**-8,932,227**	**-5,954,818**	-9,497,923	-563,285	-2,665,629	-1,812,940	0	-1,606,944	-6,757,999	-12,726,837	-14,539,777	-22,904,720
Net Salvage %	**-1930**	**-1930**	-1044	-274	-71499	-2903	0	-916	-1622	-1138	-1231	-1292

ACCOUNT 380.02 **Services - Steel**

			YEAR							3 yr. band Total 2016-2018	4 yr. band Total 2015-2018	7 yr. band Total 2012-2018
	2019 - Projected Full Yr	2019 - as of Aug 31st	2018	2017	2016	2015	2014	2013	2012			
Retirements	**1,981,341**	**1,320,894**	1,346,878	749,166	1,354,862	28	1,796	0	955,375	3,450,906	3,450,934	4,408,106
Gross Salvage	**0**	**0**	0	0	0	0	0	0	0	0	0	0
Cost to Retire	**21,305,488**	**14,203,659**	6,804,810	182,804	-48,950	0	315,066	0	24,739	6,938,664	6,938,664	7,278,469
Net Salvage$	**-21,305,488**	**-14,203,659**	-6,804,810	-182,804	48,950	0	-315,066	0	-24,739	-6,938,664	-6,938,664	-7,278,469
Net Salvage %	**-1075**	**-1075**	-505	-24	4	0	-17540	0	-3	-201	-201	-165

ACCOUNT 380.05 **Services - Copper**

			YEAR							3 yr. band Total 2016-2018	4 yr. band Total 2015-2018	7 yr. band Total 2012-2018
	2019 - Projected Full Yr	2019 - as of Aug 31st	2018	2017	2016	2015	2014	2013	2012			
Retirements	**88,436**	**58,957**	61,610	84,423	0	0	0	0	359,097	146,033	146,033	505,130
Gross Salvage	**0**	**0**	0	0	0	0	0	0	0	0	0	0
Cost to Retire	**520,509**	**347,006**	267,571	188,219	0	0	0	0	0	455,790	455,790	455,790
Net Salvage$	**-520,509**	**-347,006**	-267,571	-188,219	0	0	0	0	0	-455,790	-455,790	-455,790
Net Salvage %	**-589**	**-589**	-434	-223	0	0	0	0	0	-312	-312	-90

Exhibit__(PMN-2R)
Page 1 of 3

NATIONAL GRID - KEDLI
LPP PROGRAM
SALVAGE AND COST OF REMOVAL ANALYSIS

ACCOUNT 367.09 Mains - Steel

	2019 - Projected Full Yr	2019 - as of Aug 31st	2018	2017	2016	2015	2014	2013	2012	3 yr. band Total 2016-2018	4 yr. band Total 2015-2018	7 yr. band Total 2012-2018
							YEAR					
Retirements	0	0	0	0	0	0	37	0	73	0	0	110
Gross Salvage	0	0	0	0	0	0	0	0	0	0	0	0
Cost to Retire	0	0	0	0	0	0	883	0	531	0	0	1,414
Net Salvage$	0	0	0	0	0	0	-883	0	-531	0	0	-1,414
Net Salvage %	0	0	0	0	0	0	-2386	0	-727	0	0	-1285

ACCOUNT 376.08 Mains - Steel

	2019 - Projected Full Yr	2019 - as of Aug 31st	2018	2017	2016	2015	2014	2013	2012	3 yr. band Total 2016-2018	4 yr. band Total 2015-2018	7 yr. band Total 2012-2018
							YEAR					
Retirements	1,530,271	1,020,181	1,056,529	1,143,682	925,464	661,825	498,422	0	322,270	3,125,674	3,787,499	4,608,191
Gross Salvage	0	0	0	0	0	0	0	0	0	0	0	0
Cost to Retire	15,970,697	10,647,132	18,727,922	3,437,622	3,303,665	1,914,247	462,319	0	5,558,731	25,469,209	27,383,456	33,404,505
Net Salvage$	-15,970,697	-10,647,132	-18,727,922	-3,437,622	-3,303,665	-1,914,247	-462,319	0	-5,558,731	-25,469,209	-27,383,456	-33,404,505
Net Salvage %	-1044	-1044	-1773	-301	-357	-289	-93	0	-1725	-815	-723	-725

ACCOUNT 376.10 Mains - Cast Iron

	2019 - Projected Full Yr	2019 - as of Aug 31st	2018	2017	2016	2015	2014	2013	2012	3 yr. band Total 2016-2018	4 yr. band Total 2015-2018	7 yr. band Total 2012-2018
							YEAR					
Retirements	283,650	189,100	63,709	1	21,423	51,660	28,874	0	92,310	85,134	136,794	257,978
Gross Salvage	0	0	0	0	0	0	0	0	0	0	0	0
Cost to Retire	2,998,753	1,999,169	336,650	7,682	707,924	295,914	0	0	1,175,865	1,052,256	1,348,171	2,524,035
Net Salvage$	-2,998,753	-1,999,169	-336,650	-7,682	-707,924	-295,914	0	0	-1,175,865	-1,052,256	-1,348,171	-2,524,035
Net Salvage %	-1057	-1057	-528	-768194	-3304	-573	0	0	-1274	-1236	-986	-978

ACCOUNT 376.18 Mains - Plastic

	2019 - Projected Full Yr	2019 - as of Aug 31st	2018	2017	2016	2015	2014	2013	2012	3 yr. band Total 2016-2018	4 yr. band Total 2015-2018	7 yr. band Total 2012-2018
							YEAR					
Retirements	230,548	153,699	91,363	2,613	8,140	11,463	105,852	228	82,064	102,116	113,579	301,723
Gross Salvage	0	0	0	0	0	0	0	0	0	0	0	0
Cost to Retire	126,432	84,288	64,228	178,075	68,864	227,152	178,336	194,174	95,994	311,167	538,319	1,006,823
Net Salvage$	-126,432	-84,288	-64,228	-178,075	-68,864	-227,152	-178,336	-194,174	-95,994	-311,167	-538,319	-1,006,823
Net Salvage %	-55	-55	-70	-6816	-846	-1982	-168	-85242	-117	-305	-474	-334

ACCOUNT 380.06 Services - Plastic

	2019 - Projected Full Yr	2019 - as of Aug 31st	2018	2017	2016	2015	2014	2013	2012	3 yr. band Total 2016-2018	4 yr. band Total 2015-2018	7 yr. band Total 2012-2018
							YEAR					
Retirements	1,237,643	825,095	1,043,027	1,384,704	1,437,413	930,108	1,177,381	37,105	303,546	3,865,144	4,795,252	6,313,284
Gross Salvage	0	0	0	0	0	0	0	0	0	0	0	0
Cost to Retire	1,873,777	1,249,185	475,098	176,016	735,161	301,347	778,234	478,723	729,615	1,386,274	1,687,621	3,674,193
Net Salvage$	-1,873,777	-1,249,185	-475,098	-176,016	-735,161	-301,347	-778,234	-478,723	-729,615	-1,386,274	-1,687,621	-3,674,193
Net Salvage %	-151	-151	-46	-13	-51	-32	-66	-1290	-240	-36	-35	-58

ACCOUNT 380.01 Services - Steel

	2019 - Projected Full Yr	2019 - as of Aug 31st	2018	2017	2016	2015	2014	2013	2012	3 yr. band Total 2016-2018	4 yr. band Total 2015-2018	7 yr. band Total 2012-2018
							YEAR					
Retirements	188,526	125,684	207,958	335,587	349,393	428,396	395,552	80,916	776,103	892,938	1,321,334	2,573,905
Gross Salvage	0	0	0	0	0	0	0	0	0	0	0	0
Cost to Retire	1,735,116	1,156,744	1,300,072	1,797,309	1,641,593	472,374	4,604,948	3,227,061	3,135,816	4,738,974	5,211,348	16,179,173
Net Salvage$	-1,735,116	-1,156,744	-1,300,072	-1,797,309	-1,641,593	-472,374	-4,604,948	-3,227,061	-3,135,816	-4,738,974	-5,211,348	-16,179,173
Net Salvage %	-920	-920	-625	-536	-470	-110	-1164	-3988	-404	-531	-394	-629

ACCOUNT 380.03 Services - Copper

	2019 - Projected Full Yr	2019 - as of Aug 31st	2018	2017	2016	2015	2014	2013	2012	3 yr. band Total 2016-2018	4 yr. band Total 2015-2018	7 yr. band Total 2012-2018
					YEAR							
Retirements	18,582	12,388	22,484	24,695	21,888	8,083	34,793	1,348	28,435	69,066	77,150	141,725
Gross Salvage	0	0	0	0	0	0	0	0	0	0	0	0
Cost to Retire	43,137	28,758	12,138	10,154	2,372	297	14,744	12,463	66,586	24,664	24,962	118,755
Net Salvage$	-43,137	-28,758	-12,138	-10,154	-2,372	-297	-14,744	-12,463	-66,586	-24,664	-24,962	-118,755
Net Salvage %	-232	-232	-54	-41	-11	-4	-42	-925	-234	-36	-32	-84

Exhibit__(PMN-2R)
Page 2 of 3

KEDNY

Summary Data as of August 31, 2019

(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Account	Overall Retirements	LPP Retirements	Projected Total 2019 LPP Retirements	2018 LPP Retirements	Total Cost of Removal $	% Retirments LPP	(f) x(g) = LPP COR	Projected Total 2019 LPP COR	2018 LPP LPP COR
37605-NYGASD-T&D MAINS CAST IRON D	318,291	318,291	**477,436**	**672,585**	15,869,155	100.00%	15,869,155	**23,803,733**	**22,887,533**
37609-NYGASD-T&D MAINS STEEL DISTR	1,172,159	308,472	**462,708**	**909,583**	7,951,249	74.89%	5,954,818	**8,932,227**	**9,497,923**
38000-NYGASD-T&D GAS SERVICES MIXED	-	-	-	-	(13,701)	0.00%	-	-	-
38002-NYGASD-T&D GAS SERVICES STL P	9,815,471	1,320,894	**1,981,341**	**1,346,878**	20,011,727	70.98%	14,203,659	**21,305,488**	**6,804,810**
38005-NYGASD-T&D GAS SERV CAST IRON	58,957	58,957	**88,436**	**61,610**	347,006	100.00%	347,006	**520,509**	**267,571**
Total			**3,009,921**	**2,990,656**				**54,561,958**	**39,457,837**

KEDLI

Summary Data as of August 31, 2019

Account	Overall Retirements	LPP Retirements	Projected Total 2019 LPP Retirements	2018 LPP Retirements	Total Cost of Removal $	% Retirments LPP	(f) x(g) = LPP COR	Projected Total 2019 LPP COR	2018 LPP LPP COR
36709 - NYGASD-MAINS STEEL TRANSM >92	351	-	-	-	18,963.83	0.00%	-	-	-
37608 - NYGASD-T&D MAINS STEEL	2,215,465	1,020,181	**1,530,271**	**1,056,529**	11,563,068.02	92.08%	10,647,132	**15,970,697**	**18,727,922**
37610-NYGASD-T&D MAINS CAST IRON D	189,100	189,100	**283,650**	**63,709**	1,999,168.67	100.00%	1,999,169	**2,998,753**	**336,650**
37618-NYGASD-T&D MAINS PLASTIC DIST	6,295,124	153,699	**230,548**	**91,363**	809,576.94	10.41%	84,288	**126,432**	**64,228**
38001-NYGASD-T&D GAS SERVICES STEEL	207,517	125,684	**188,526**	**207,958**	1,207,563.52	95.79%	1,156,744	**1,735,116**	**1,300,072**
38003-NYGASD-T&D GAS SVS COPPER >92	12,388	12,388	**18,582**	**22,484**	28,758.33	100.00%	28,758	**43,137**	**12,138**
38006-NYGASD-T&D GAS SERVICES PLAST	17,627,232	825,095	**1,237,643**	**1,043,027**	6,974,445.48	17.91%	1,249,185	**1,873,777**	**475,098**
Total			**3,489,219**	**2,485,070**				**22,747,913**	**20,916,108**

***KEDNY Account 38000 Contorl was broken up by material type and assets were transferred to the 380xx sub-accounts

Exhibit__(PMN-2R)
Page 3 of 3